

Supplementary Tables S1–S7

Gamma-Code Consensus NoC Submission

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This file contains detailed numerical tables moved out of the main manuscript to keep the main NoC submission within length and readability constraints.

Table 1: Supplementary Table. Retuning vs. re-entry effect across four baseline coupling strengths ($N_{\text{seeds}} = 20$ per K). Retuning effect: difference $\langle \Pi_{\text{full}}^A \rangle(\text{decline}) - \langle \Pi_{\text{full}}^A \rangle(\text{baseline})$. Re-entry effect: difference $\langle \Pi_S^B \rangle - \langle \Pi_S^A \rangle$ in phase 3.

K_{baseline}	Retuning $\Delta\Pi$	$\langle \Pi_S^A \rangle$	$\langle \Pi_S^B \rangle$	B–A	Rel. increase	Wilcoxon p
1.6	+0.16	0.15	0.27	+0.12	+81%	0.006
2.0	+0.07	0.26	0.53	+0.27	+102%	0.0009
2.5	+0.19	0.52	0.75	+0.23	+44%	0.004
3.0	+0.15	0.73	0.76	+0.04	+5%	0.54 n.s.

Table 2: Supplementary Table. Access indicator Π per sedation level and paired Wilcoxon signed-rank tests (baseline vs. moderate). $N = 20$ participants.

Band	Π_{base}	Π_{mild}	Π_{mod}	Π_{rec}	$\Delta\Pi$ (base–mod)	p -value
Gamma	0.661	0.604	0.543	0.569	+0.118	6.3×10^{-5}
Alpha	0.659	0.641	0.561	0.632	+0.098	0.0037

Table 3: Supplementary Table. Frozen cross-dataset benchmark. AUC and balanced accuracy for selected no-fit and train-on-one-dataset/test-on-the-other models. Sign-rule models use no fitted parameters. Logistic transfer models fit scaling and weights only on the training dataset.

Model	Direction	Band	AUC	Balanced accuracy
Standard Π sign rule	pooled no-fit	gamma	0.967	0.983
Standard Π delta	Chennu→DS005620	gamma	1.000	1.000
Standard Π delta	DS005620→Chennu	gamma	0.900	0.725
wPLI Π sign rule	pooled no-fit	gamma	0.858	0.925
wPLI triad + Π	Chennu→DS005620	gamma	0.950	0.950
wPLI triad + Π	DS005620→Chennu	gamma	0.900	0.825
Spectral bandpowers	Chennu→DS005620	gamma	0.975	0.938
Spectral bandpowers	DS005620→Chennu	gamma	0.900	0.775

Table 4: Supplementary Table. Sleep-EDF sigma-band state differentiation under leave-recording-out validation. Values are computed on 22 Sleep-EDF Sleep Cassette recordings. Wake and REM are each compared against NREM. Spectral features remain stronger for canonical sleep staging; GCC is interpreted as cross-paradigm state geometry rather than as a replacement sleep-stage biomarker.

Model	Wake vs. NREM AUC	REM vs. NREM AUC
Spectral features	0.959	0.892
GCC triad + Π	0.875	0.727
Π only	0.534	0.534
Residual GCC after spectral regression	0.637	0.572

Table 5: Supplementary Table. Additional robustness and boundary-domain analyses. These analyses are not used to claim bandpower independence or direct P1 validation; they test implementation stability and empirical breadth.

Analysis	Main result	Interpretation
Amplitude-normalized phase-only GCC	Chennu gamma $\Delta\Pi = 0.463$; DS005620 gamma awake-vs-sed $\Delta\Pi = 0.439$	Propofol sensitivity is not solely produced by amplitude entering the GCC observables.
CSD/source-proxy phase-only GCC	Chennu gamma $\Delta\Pi = 0.159$; DS005620 gamma awake-vs-sed $\Delta\Pi = 0.393$	Effect survives a surface-Laplacian source-proxy; individualized MRI source reconstruction remains future work.
Subanaesthetic ketamine	Gamma Π shift -0.142 ; alpha D_{eff} shift $+0.631$	Altered-state shift, not propofol-like collapse.
DS004504 dementia proxy	Dementia-vs-control combined triad AUC = 0.769	Degradation sensitivity, not direct access validation.
Reliability	Phase-only/CSD split-half gamma Π ICC ≈ 0.92 –0.93; repeated-run gamma ICC ≈ 0.67 –0.87	GCC Π is stable enough for recording-level analysis, with lower reliability across distinct acquisition runs.

Table 6: Supplementary Table. P3/P4 Ensemble ($N_{\text{seeds}} = 10$ per cell): shift of $\langle K_c^{\text{op}} \rangle$ and maximum access index $\langle \max_K \Pi(K) \rangle$ with increasing degradation. Values are means $\pm$ standard deviations over the seed ensemble. Both quantities behave monotonically under increasing lesion.

Lesion f	$\langle K_c^{\text{op}} \rangle \pm \text{SD}$	$\langle \max_K \Pi(K) \rangle \pm \text{SD}$
0.00	1.38 ± 0.57	0.83 ± 0.29
0.15	1.49 ± 0.44	0.81 ± 0.24
0.30	1.80 ± 0.40	0.71 ± 0.30
0.45	2.15 ± 0.33	0.58 ± 0.38

Table 7: Supplementary Table. P6: Phase-wise access indices Π (fraction of time inside the access region), averaged across $N_{\text{seeds}} = 20$ independent seeds (mean $\pm$ SD). **Critical finding:** In phase 3 (re-entry window), condition B (selective preservation) shows a significantly higher access index on backbone S than condition A (uniform): $\langle \Pi_S^B \rangle = 0.273$ vs. $\langle \Pi_S^A \rangle = 0.151$. In phase 4, both collapse nearly completely, with a marginal residual signal in condition B ($\langle \Pi_S^B \rangle = 0.07$) vs. condition A ($\langle \Pi_S^A \rangle = 0.00$) consistent with a short-lived gain residuum on S ; the re-entry is thus essentially transient.

Phase	Π A full	Π B full	Π A on S	Π B on S
Phase 1 baseline	0.53 ± 0.38	0.53 ± 0.38	0.54 ± 0.38	0.54 ± 0.38
Phase 2 decline	0.69 ± 0.33	0.69 ± 0.33	0.61 ± 0.33	0.61 ± 0.33
Phase 3 re-entry	0.09 ± 0.12	0.09 ± 0.13	0.15 ± 0.19	0.27 ± 0.27
Phase 4 final	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.07 ± 0.11